

EDUCATION

Northwestern University — *Evanston, IL*

Masters of Science in Robotics December 2025

University of Wisconsin-Madison — *Madison, WI*

Bachelor of Arts in Computer Science December 2022

SKILLS

Embedded Systems & Firmware: C, C++, Assembly, Embedded C, PIC32, Jetson, Circuit Analysis, PCB Design

Hardware & Robotics: ROS2, Gazebo, MoveIt, Nav2, OpenCV, Robot Kinematics, RViz, Oscilloscope, Soldering

Software Development: Linux, Git, Python, Bash, Docker, Swift, Kotlin, React.js, Django, Node.js

DevOps & Infrastructure: AWS (RDS, EKS, EC2, Lambda), Kubernetes, Jenkins, Prometheus, Ansible

Machine Learning & Vision: PyTorch, TensorFlow, CUDA, YOLO, FFTs, Transformers

WORK EXPERIENCE

EL Tracker — el-tracker.com

October 2023 – Present

Founder and Sole Developer

Chicago, IL

- Built a **real-time transit tracking app** integrating computer vision, geolocation, and accessibility features
- Developed **low-latency backend services** in Django and Node.js, with mobile clients in Swift & Kotlin
- Engineered robust infrastructure with **AWS, Docker, Kubernetes**, and a **real-time WebSocket API**
- Achieved **3,000+ downloads** and a **5-star rating**, ranking #46 in the nationwide navigation category

Sunrise Futures

January 2023 – January 2024

Senior DevOps Associate

Chicago, IL

- Rebuilt a **low-latency monitoring system** in **C++ and SQL**, improving system performance and cutting costs
- Conducted **daily testing of trading algorithms**, deploying updates via **Docker and Ansible**.
- Mentored junior developers, focusing on **infrastructure troubleshooting and performance optimization**

Datachat

January 2022 – January 2023

Software Engineer

Madison, WI

- Integrated clustering models, such as K-means, in a data analytics platform via PyTorch, BigQuery
- Deployed features via React and Go to clients such as Meta, by integrating feedback from weekly syncs

Leverage Lab

June 2021 – August 2021

Data Engineering Intern

Kansas City, MO

- Developed Google Cloud Pub/Sub for real-time data streaming and messaging between cloud components
- Automated **data validation scripts**, improving ingestion accuracy across cloud components

FEATURED PROJECTS

Quadrotor Flight Controller from Scratch

April 2025 – Present

- Created **real-time** embedded C flight-control for **IMU calibration**, sensor filtering, and I²C attitude estimation
- Tuned **multi-axis PID** and navigation algorithms for position hold, waypoint tracking, and **yaw control**

Gesture-Controlled Robot with AR Glasses & ROS2

December 2024 – March 2025

- Built a **C++ ROS2 package** enabling **gesture-based robot control** via **MediaPipe and AR Glasses**
- Designed an interface for **Unitree Go1 & Franka Emika Panda arm**, enabling **real-time motion control**

Robot Whack-a-mole Player

November 2024 – December 2024

- Programmed a **7-DOF robotic arm** using **ROS2 MoveIt** to control a servo-driven hammer
- Applied **OpenCV & YOLOv8** to detect mole illumination events with **millisecond-level reaction time**

PERSONAL

Professional Chess Player

Apr 2009 – Present

- Achieved 2100 USCF (United States Chess Federation) Rating, ranking top 20 in Wisconsin
- Earned \$6,500 by winning the Under-2000 Category at the World Open 2022 Chess Tournament